

BMJ Open REenergizeME: intermittent hypoxia-hyperoxia treatment for myalgic encephalomyelitis/chronic fatigue syndrome – protocol for a randomised, placebo-controlled trial

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To cite: Nochi M, Kristensen SK, Mehrani N, *et al.* REenergizeME: intermittent hypoxia-hyperoxia treatment for myalgic encephalomyelitis/chronic fatigue syndrome—protocol for a randomised, placebo-controlled trial. *BMJ Open* 2026;**16**:e117729. doi:10.1136/bmjopen-2026-117729

► Prepublication history and additional supplemental material for this paper are available online. To view these files, please visit the journal online (<https://doi.org/10.1136/bmjopen-2026-117729>).

Received 01 February 2026
Accepted 24 July 2026



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ABSTRACT

Introduction Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) is a chronic, disabling condition characterised by post-exertional malaise (PEM), autonomic dysfunction and markedly reduced quality of life. Its prevalence has increased following the COVID-19 pandemic, yet no effective disease-modifying treatments are available and underlying mechanisms remain unclear. Evidence implicates immune-metabolic dysregulation, impaired energy metabolism, vascular dysfunction and altered oxygen handling. Intermittent hypoxia-hyperoxia treatment (IHHT) is a non-pharmacological intervention proposed to induce adaptive physiological and cellular responses via hypoxia- and redox-sensitive pathways. This trial evaluates the clinical efficacy of IHHT in ME/CFS and explores associated mechanisms.

Methods and analysis This randomised, placebo-controlled, participant- and assessor-blinded clinical trial will enrol 104 female patients aged 20–59 years who meet the International Consensus Criteria for ME/CFS along with a healthy control group for mechanistic comparisons. Participants will be randomised 1:1 to IHHT or placebo following baseline assessments. The intervention will be delivered over 8 weeks. The primary endpoint is change in quality of life measured by the vitality domain of the Short Form-36 Health Survey. Secondary endpoints include PEM severity, fatigue severity, cognitive severity, functional capacity and autonomic symptoms assessed using validated patient-reported outcome measures and objective tests. Exploratory endpoints include pain phenotyping, autonomic, sudomotor and neurophysiological assessments, tissue oxygenation by near-infrared spectroscopy, peripheral small-fibre nerve assessments using corneal confocal microscopy and skin biopsy and immune-metabolic and oxidative stress biomarkers. Assessments will be conducted at baseline; post-treatment; and at 3, 6 and 12 months follow-up.

STRENGTHS AND LIMITATIONS OF THIS STUDY

- ⇒ Randomised, placebo-controlled, participant- and assessor-blinded clinical trial conducted in compliance with International Council for Harmonisation – Good Clinical Practice guidelines.
- ⇒ Comprehensive assessment strategy combining patient-reported outcomes with objective physiological, neurobiological and molecular measures, including autonomic testing, tissue oxygenation, peripheral small-fibre nerve assessments and immune-metabolic profiling.
- ⇒ The intervention protocol and sample size are based on prior experience and data from related post-viral fatigue conditions rather than on a dedicated pilot trial in myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS).
- ⇒ Participants are recruited from specialised clinical settings, which may limit generalisability to individuals with milder disease or shorter symptom duration or those with severe disease who are bedbound and therefore unable to participate in the study procedures.
- ⇒ The inclusion of female participants only reflects the higher prevalence of ME/CFS in women and improves cohort homogeneity but limits the generalisability of the findings to male patients.

Ethics and dissemination The trial is conducted in accordance with the Declaration of Helsinki and International Council for Harmonisation – Good Clinical Practice guidelines and was approved by the Danish Scientific Ethical Committees (De Videnskabetiske Medicinske Komitéer, VMK; approval no. 2500959). Written informed consent will be obtained from all participants. Results will be disseminated through peer-reviewed publications and scientific conferences.

Trial registration number ClinicalTrials.gov Identifier: [NCT07317401](https://clinicaltrials.gov/ct2/show/study/NCT07317401).

INTRODUCTION

Myalgic encephalomyelitis (ME), also referred to as chronic fatigue syndrome (CFS), is a chronic, debilitating neurological disorder in which post-exertional malaise (PEM) is the cardinal feature. PEM is defined as a prolonged worsening of symptoms—including cognitive and physical fatigue, pain and autonomic disturbances—triggered by minimal physical, cognitive or orthostatic stress.^{1 2} Globally, ME/CFS is estimated to affect approximately 0.5%–1.0% of the population, corresponding to tens of millions of individuals worldwide.^{3–5} In Denmark, approximately 16000 individuals are diagnosed with the condition, and patients experience greater functional impairment than those with depression, multiple sclerosis or stroke.⁶ Importantly, population-based studies following the COVID-19 pandemic indicate that the risk of developing ME/CFS or ME/CFS-like illness has increased substantially after SARS-CoV-2 infection, with estimates suggesting up to a fivefold higher incidence compared with non-infected populations.^{7–9} Despite its substantial personal, social and economic impact, no effective treatment is currently available, and the underlying pathophysiology remains poorly understood.

The post-infectious syndrome known as long COVID shares substantial clinical and biological overlap with ME/CFS, including PEM, fatigue, myalgia, cognitive impairment ('brain fog') and autonomic disturbances.^{10–12} Increasing evidence indicates that both conditions share a post-infectious, immune-mediated pathophysiology and are characterised by vascular, immune, metabolic and neuropathic dysfunction domains. Although the drivers of these interacting pathophysiological domains remain unclear, persistent immune activation, chronic viral reactivation, autoimmunity and mitochondrially controlled inflammatory and redox abnormalities have been proposed as potential contributors.^{10–12}

Functional autoantibodies targeting adrenergic, muscarinic and endothelin receptors have been identified in a significant proportion of patients. These autoantibodies may disrupt vascular tone, microcirculatory perfusion and autonomic signalling, thereby contributing to impaired cardiovascular regulation and altered control of oxygen delivery and utilisation within the neurovascular and neuromuscular microenvironment.^{13–20}

The resulting hypoxic environment may induce mitochondrial remodelling, reduce metabolic flexibility and exacerbate oxidative stress and inflammation. Oxidative damage, including lipid peroxidation, further impairs endothelial function, red blood cell deformability and microvascular perfusion, all of which have been observed in patients with ME/CFS.^{15 21–25} Periodic disturbances in oxygen delivery to brain and muscle tissues may therefore trigger recurrent episodes of mild hypoxic injury, mitochondrial dysfunction, oxidative stress and inflammation, collectively contributing to PEM. This mechanism may explain why even minimal activity above an individual's threshold provokes disproportionate fatigue and pain in ME/CFS. Despite heterogeneity in specific

pathophysiological markers, evidence of oxidative stress, mitochondrial metabolic alterations, microvascular changes, impaired blood flow and autonomic dysfunction in ME/CFS, and more recently in long COVID, is abundant and shows a recurring pattern.^{1 8 11 12 15 21 22 24–31}

Given the rising prevalence of long COVID and the chronic, disabling nature of ME/CFS, the associated public health and economic burden is expected to increase substantially, underscoring the urgent need for novel therapeutic strategies.³² Intermittent hypoxia-hyperoxia treatment (IHHT) represents a promising approach by directly targeting several of the pathophysiological mechanisms implicated in ME/CFS. IHHT involves repeated cycles of hypoxic and hyperoxic exposure, which enhance hypoxia tolerance through activation of redox-sensitive cellular stress-response pathways, including mitochondrial antioxidant and repair pathways to improve mitochondrial function and reduce oxidative and inflammatory stress.^{33–35} At mild, hormetic levels, hypoxia stabilises hypoxia-inducible factor 1 alpha, leading to the transcription of genes involved in erythropoiesis, angiogenesis, vascular blood flow regulation and metabolism, thereby promoting physiological and cellular adaptation to hypoxic stress.^{34 35 37 38} Two studies applying IHHT and one study using intermittent hypoxia treatment in patients with long COVID have reported favourable safety profiles and improvements in self-reported fatigue, pain and functional capacity, along with improvements in exercise performance, inflammatory markers and autonomic function in non-placebo-controlled patient cohorts.^{39–41} We hypothesise that IHHT preconditions brain and muscle tissues to tolerate recurrent mild hypoxic stress, thereby improving PEM, autonomic regulation of cardiovascular and metabolic energy function and overall quality of life in patients with ME/CFS. Furthermore, we expect that clinical and functional improvement will correspond with enhanced microvascular function and changes in systemic immune–metabolic and oxidative stress biomarkers. Accordingly, the objective of this trial is to evaluate the clinical efficacy of IHHT in ME/CFS and to explore its effects on autonomic function and underlying immune–metabolic, vascular and neuropathic mechanisms. The broader perspective is to generate new hypotheses about ME/CFS disease mechanisms and, if the trial demonstrates positive effects, to inform the development of future IHHT protocols. The primary endpoint is quality of life, specifically the vitality domain of the self-reported Short Form-36 Health Survey (SF-36) questionnaire, which focuses on energy level and fatigue. Secondary endpoints include other measures of clinical improvement, such as the physical role domain of the SF-36, PEM severity (assessed using the Functional Capacity (FUNCAP) questionnaire alongside cognitive and hand-grip strength tests), autonomic function (assessed using the Composite Autonomic Symptom Score-31 (COMPASS-31) questionnaire, electrocardiographic data and heart-rate variability during orthostatic stress testing and Valsalva manoeuvre) and pain severity

(assessed using the painDETECT questionnaire and detailed pain assessment - including pain mapping and sensory examination). Explorative endpoints comprise complementary assessments of underlying disease mechanisms to identify potential biological and functional response patterns. These include corneal confocal microscopy (CCM) and skin biopsy to quantify nerve-fibre density, macrophages, mitochondria and vascular structures, tissue oxygenation assessed by near-infrared spectroscopy (NIRS) and cellular and plasma biomarkers, to capture mitochondrial, vascular and oxidative/inflammatory stress markers.

METHODS AND ANALYSIS

Trial design

This study is a prospective, randomised, placebo-controlled, participant- and assessor-blinded clinical trial. Participant recruitment is planned to commence in June 2026, with final follow-up and completion of data collection anticipated in January 2029. The study comprises a screening visit, a baseline assessment visit, six treatment sessions and one post-treatment follow-up visit.

During screening, participants will be evaluated for eligibility according to predefined inclusion and exclusion criteria. Individuals meeting all criteria will proceed to the baseline assessment. If temporary exclusion criteria are identified, participants may still be included provided these are re-evaluated and resolved before treatment initiation. The maximum interval between screening and treatment commencement is 7 weeks.

Eligible participants will be randomised in a 1:1 ratio to receive either an IHHT protocol or a sham protocol. Block randomisation will be used, stratified by age, disease duration and disease severity, classified according to the National Institute for Health and Care Excellence (NICE) guidelines for ME/CFS.

Participants will receive six treatment sessions over 8 weeks. Treatments will be administered by trained staff members who are not involved in outcome assessment, data collection or data analysis. Outcome assessors and investigators responsible for data analysis will remain

blinded to treatment allocation. Participants will be blinded to treatment allocation throughout the study.

To minimise bias, treatment staff will not be involved in clinical assessments and will refrain from discussing treatment allocation or hypotheses with participants.

The success of blinding will be evaluated after completion of the intervention. Participants will be asked to indicate whether they believe they received active IHHT or sham treatment and to specify the reason for their belief (eg, perceived effects, side effects or other cues such as sensory experiences or staff interaction).

Detailed procedures for randomisation, allocation concealment and blinding are defined in the study protocol and operational procedures.

The post-treatment follow-up visit will take place no earlier than 2 weeks after the final treatment session. Participants will additionally be invited to complete online self-reported follow-up questionnaires at 3, 6 and 12 months post-intervention.

A schematic overview of the trial design is shown in figure 1.

Patient and public involvement

The trial design was discussed with and informed by a patient advisory board, including representatives from the Danish ME Association (ME Foreningen). The advisory board provided input on the relevance and feasibility of the study design.

Setting

This investigator-initiated trial will be conducted at Aarhus University Hospital in collaboration with Remedical.Me ApS and SANA Medical Systems, all located in Denmark. The trial sponsor is Aarhus University, in partnership with the Central Denmark Region represented by Aarhus University Hospital, Aarhus, Denmark.

Patients diagnosed with ME/CFS will be identified through the patient registry at Remedical.Me ApS, and eligible individuals will be contacted via email. Additional participants will be recruited through advertisements distributed via the Danish ME Association (ME Foreningen), the Danish COVID Association (Dansk

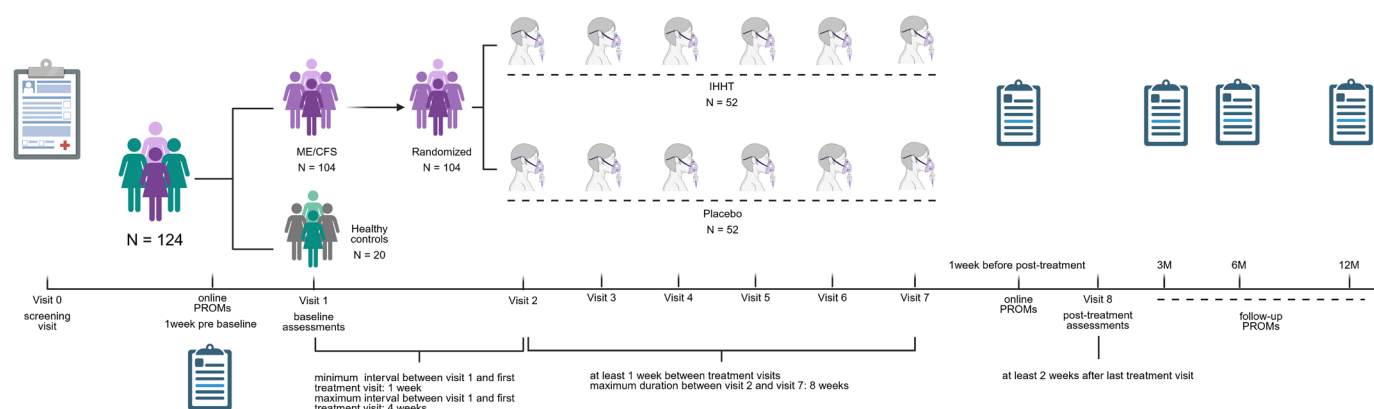


Figure 1 Trial flowchart.

Covidforening) and social media platforms. Healthy controls will be recruited through public advertisements, including posts on social media (eg, Facebook and LinkedIn), the Danish ME Association's and Danish COVID Association's websites, and local networks at Aarhus University Hospital.

Baseline and post-treatment assessments will be conducted at Aarhus University Hospital, while treatment sessions will be administered at SANA Medical Systems' HypoxBreath facilities in Risskov and Copenhagen. All participants, whether in the intervention or control group, will receive treatment under the supervision of intensivists and nurses specially trained in IHHT procedures.

The funders had no role in the design of the study; collection, analysis or interpretation of data; writing of the manuscript; or the decision to submit the protocol for publication.

Trial population

The study will recruit at least 104 female patients aged 20–59 years who meet the International Consensus Criteria for ME/CFS.⁴² Eligible participants must have a disease duration exceeding 6 months and a documented history of PEM. Only individuals classified with mild to moderate disease severity according to the NICE guidelines⁴³ will be included.

Participants with other clinically significant chronic psychiatric, neurological, cardiovascular, metabolic or systemic diseases not considered part of their ME/CFS will be excluded, in order to minimise confounding and ensure participant safety. Additional exclusion criteria include pregnancy, obstructive lung disease, claustrophobia, a history of spontaneous pneumothorax or the presence of any medical device or condition that may interfere with IHHT.

A control group of at least 20 healthy individuals, matched for age, sex, body mass index (BMI) and activity level, will be included for baseline comparisons. Controls must be in good general health, without chronic pain, neurological or psychiatric disorders and will complete the same baseline assessments as patients.

Inclusion and exclusion criteria for both patients and controls are detailed in [table 1](#) and [table 2](#). At the start of each of the six treatment visits, subjects will be excluded from the period if any of the criteria listed in [table 3](#) apply. All patients will undergo a comprehensive assessment, including questionnaires, physical examinations and laboratory testing at baseline and post-treatment.

Treatment/intervention

Participants in the IHHT group will receive therapy using the HypoxBreath device (Telmed Telemedizin GmbH, Germany; CE-marked medical device, class IIa). The HypoxBreath system integrates a built-in compressor, an air reservoir and a membrane module that dynamically adjusts the fraction of inspired oxygen (FiO₂) by adding or removing oxygen from ambient air. Treatment

parameters are controlled via a user interface connected to a monitoring system and data-logging server.

Each treatment session consists of four to seven alternating hypoxia–hyperoxia cycles, lasting a total of 22–40 min. During therapy, arterial oxygen saturation (SpO₂) fluctuates cyclically due to variable FiO₂ levels, ranging from 9% to 13% during the hypoxic phase to 36% during the hyperoxic restitution phase. Heart rate, SpO₂, respiratory rate, heart rate variability and FiO₂ are continuously monitored throughout each session.

Treatment settings are individually calibrated based on physiological parameters and participant tolerance. Adjustments may be made during the session to account for relaxation level, anxiety or physiological fatigue. The clinical protocol applied at SANA Medical Systems is based on data from more than 3000 patients treated with IHHT.

The selected treatment frequency and interval were based on clinical experience, feasibility considerations and safety considerations specific to individuals with ME/CFS, particularly the risk of symptom exacerbation and PEM. Unlike more intensive IHHT regimens administered several times weekly in rehabilitation settings, the present protocol applies a lower-frequency, individualised dosing strategy intended to allow sufficient physiological recovery and adaptive responses between sessions. This approach is based on the principle of hormesis, whereby controlled intermittent hypoxia exposure may stimulate adaptive mitochondrial, autonomic, vascular and cellular stress-response pathways while minimising excessive physiological burden. Previous observational data from patients with post-COVID-19 condition demonstrated clinically meaningful improvements in fatigue, pain and quality of life following a median of six individualised IHHT sessions.⁴⁰

Participants in the placebo (sham) group will undergo simulated IHHT sessions using room air (21% FiO₂). To mimic the auditory and mechanical sensations of IHHT, the HypoxBreath device will reproduce airflow fluctuations corresponding to the treatment intervals. The sham intervention is designed to mimic the procedural aspects of IHHT, including airflow delivery, mask application and device-related acoustic cues, while delivering normoxic air. However, as the sham condition does not reproduce the physiological effects associated with hypoxia and hyperoxia exposure, some participants may perceive differences between treatment conditions.

Each participant will complete six treatment sessions over 8 weeks, with sessions spaced 6–8 days apart to allow physiological recovery and adaptation between exposures. Patient-centred and self-reported outcomes will be recorded at each visit prior to treatment.

Procedures

All potential participants will receive oral and written information about the study and provide written informed consent prior to any study-related procedures. An example of the participant consent form is provided in the

Table 1 Inclusion criteria at screening visit

	Patients	Controls
	Justification/rationale	Justification/rationale
Diagnosis	Established diagnosis of ME/CFS using the International Consensus Criteria ⁴² with disease duration of more than 6 months and exercise intolerance/PEM	Subjects must be in good health as determined by the medical history, physical examinations and laboratory assessments, and must not show any clinically significant deviations from reference ranges on 12-lead ECG, vital signs (blood pressure, pulse rate and respiratory rate) or laboratory parameters (renal function and electrolyte balance, haematological parameters, immune-inflammatory markers and thyroid function tests)
	To enrol a well-defined and homogeneous group of patients diagnosed with ME/CFS	Interpretability of results
Disease severity	Classification as having a mild/moderate disease severity according to the NICE guidelines ⁴³	N/A
	To enrol capable patients to meet up at the clinic for inclusion and IHHT sessions	
BMI	Body mass index >18 kg/m ² and <30 kg/m ²	Body mass index >18 kg/m ² and <30 kg/m ²
	Exclude obesity and being underweight due to its impact on mitochondrial function	Consistent with being in good health To ensure alignment with the patient group
Gender and age	Female, 20–59 years	Female, 20–59 years
	To minimise variability	To minimise variability
Contraception use	Use of highly effective contraception with a low failure rate defined as <1% per year for at least 1 month prior to screening and agreement to use such a method during study participation and for an additional 1 week after the end of IHHT:	Use of highly effective contraception with a low failure rate defined as <1% per year for at least 1 month prior to screening and agreement to use such a method during study participation:
	▶ Combined (oestrogen and progestogen containing) hormonal contraception ▶ Progestogen-only hormonal contraception associated with inhibition of ovulation ▶ An intra-uterine device (hormone-free) ▶ An intra-uterine hormone releasing system (IUS)	▶ Combined (oestrogen and progestogen containing) hormonal contraception ▶ Progestogen-only hormonal contraception associated with inhibition of ovulation ▶ An intra-uterine device (hormone-free) ▶ An IUS
	A woman of non-childbearing potential may be included if surgically sterile (ie, after hysterectomy or bilateral oophorectomy) or post-menopausal for at least 2 years.	A woman of non-childbearing potential may be included if surgically sterile (ie, after hysterectomy or bilateral oophorectomy) or post-menopausal for at least 2 years.
Informed consent	To avoid pregnancies with potential harm to the unborn	To ensure alignment with the patient group
	Provision of signed and dated informed consent form	Provision of signed and dated informed consent form
	Ethical requirement	Ethical requirement

Continued

Table 1 Continued

	Patients	Controls
	Justification/rationale	Justification/rationale
Compliance with study procedures	Stated willingness to comply with all study procedures and regimens and availability for the duration of the study Ethical requirement and to minimise dropout rate	Stated willingness to comply with all study procedures and regimens and availability for the duration of the study Ethical requirement and to minimise dropout rate
Physical activity	N/A	Not performing regular strenuous physical activity every week To match for sedentary lifestyle of the patients
IHHT, intermittent hypoxia-hyperoxia treatment; ME/CFS, myalgic encephalomyelitis/chronic fatigue syndrome; NICE, National Institute for Health and Care Excellence; PEM, post-exertional malaise.		

online supplemental material. A screening visit will then be conducted to confirm eligibility based on predefined inclusion and exclusion criteria. Demographic information, medical history and current medications will be recorded, and a safety screening for contraindications to IHHT will be completed.

Eligible participants will then attend a baseline assessment that includes a comprehensive battery of self-reported questionnaires evaluating fatigue, PEM, pain, autonomic symptoms and quality of life, together with physiological and laboratory assessments. Participants fulfilling all inclusion criteria will be randomised 1:1 to receive either IHHT using the HypoxBreath device or a sham (placebo) protocol.

Each participant will complete six treatment sessions over an 8-week period, scheduled at approximately 6–8-day intervals. A post-treatment assessment will be conducted no earlier than 2 weeks after the final session, repeating the full baseline evaluation to assess short-term effects. Participants will additionally complete online follow-up questionnaires at 3, 6 and 12 months post-intervention to assess the persistence of treatment effects.

All trial procedures will be conducted in accordance with the study's SOPs and International Council for Harmonisation – Good Clinical Practice (ICH-GCP) guidelines. A summary of study procedures and assessment time points is provided in online supplemental table 1.

Assessments/measurements

A comprehensive battery of clinical, physiological, neuro-cognitive and biochemical assessments will be undertaken to evaluate the efficacy of IHHT and to elucidate its impact on autonomic regulation, mitochondrial function and systemic pathophysiology in ME/CFS. All assessments will be standardised and performed in accordance with the study SOPs and ICH-GCP guidelines. Data will be collected at baseline (Visit 1) and post-treatment (Visit 8), unless otherwise specified. In addition, online follow-up assessments will be conducted at 3, 6 and 12 months post-intervention to evaluate the durability of treatment effects on quality of life, fatigue, pain and autonomic symptoms.

At each on-site assessment, participants will undergo a structured sequence of evaluations including patient-reported outcome measures (PROMs), physical and cognitive performance tests, autonomic and sensory profiling, tissue oxygenation assessments and biological sampling for mechanistic biomarkers.

PROMs

SF-36

The SF-36 is a 36-item self-administered questionnaire designed to assess health-related quality of life across eight domains during the preceding 4 weeks: physical functioning (10 items), role limitations due to physical health (four items), role limitations due to emotional problems (three items), energy/fatigue (four items), emotional well-being (five items), social functioning (two items), pain (two items) and general health (five items).^{44 45} One additional item evaluates perceived change in overall health compared with 1 year earlier. Each item is coded and transformed to a 0–100 scale according to the SF-36 scoring manual, and the eight domain scores are derived as the average of their transformed item scores, with higher scores indicating better perceived health status. The vitality domain, reflecting energy and fatigue, will serve as the primary endpoint in this study, as it directly captures the core symptomatology of ME/CFS. The validated Danish version of the SF-36, for which reliability, scaling assumptions and population norms have been established, will be applied.⁴⁶

FSS

The Fatigue Severity Scale (FSS) is a validated self-administered questionnaire designed to assess the severity and functional impact of fatigue. The FSS consists of nine items evaluating how fatigue interferes with motivation, physical functioning, work, family and social life. Each item is scored on a 7-point Likert scale, with higher scores indicating greater fatigue severity.^{47 48}

The FSS has demonstrated good validity, internal consistency and test–retest reliability across multiple chronic and neurological conditions, including autoimmune and

Table 2 Exclusion criteria at screening visit

	Patients	Controls
	Justification/rationale	Justification/rationale
1	Participation in other clinical trials. To avoid interference with the purpose of the study	Participation in other clinical trials. To avoid interference with the purpose of the study
2	Inability to follow the treatment plan, incl. known medical conditions that might place the patient at risk from the IHHT, or confound the interpretation of the test results, including inability to avoid or limit medication as required by the study protocol. To avoid interference with the purpose of the study Subject safety	Inability to follow the study procedures, incl. known medical conditions that might confound the interpretation of the test results, including inability to avoid or limit medication as required by the study protocol. Consistent with being in good health
3	Presence of any medical devices (eg, cardiac pacemaker), implants or prosthesis unless it is beyond discussion that these will not put the subject's safety during the study at risk and will not interfere with the results of the study. To avoid interference with the purpose of the study	Presence of any medical devices (eg, cardiac pacemaker), implants or prosthesis unless it is beyond discussion that these will not put the subject's safety during the study at risk and will not interfere with the results of the study. Consistent with being in good health
4	Any relevant symptom of neurological dysfunction of the motor and sensory system that is not part of the ME/CFS disease or may interfere with the conduct of the study. To avoid interference with the purpose of the study	Any relevant symptom of neurological dysfunction of the motor and sensory system that may interfere with the conduct of the study. Consistent with being in good health
5	History or clinically significant symptoms of psychiatric disease that are not considered part of the ME/CFS disease and that may impact the conduct or interpretation of study assessments (eg, major depressive disorder, bipolar disorder, psychotic disorders). To avoid interference with the purpose of the study	History or clinically significant symptoms of psychiatric disease that may impact the conduct or interpretation of study assessments (eg, major depressive disorder, bipolar disorder, psychotic disorders). Consistent with being in good health
6	History or symptoms of central nervous system disease or peripheral nerve lesions or dysfunction that are not part of the ME disease or that may impact the study assessments. To avoid interference with the purpose of the study	History or symptoms of central nervous system disease or peripheral nerve lesions or dysfunction with sequelae that may impact the study assessments. Consistent with being in good health
7	N/A	Not willing or able to abstain from changes in physical exercise activities during the study. To avoid interference with the purpose of the study
8	Any chronic pain condition not considered part of the ME/CFS condition or recent (ie, within the preceding 2 years) history thereof, that may impact study assessments. To avoid interference with the purpose of the study	Any chronic pain condition or recent (ie, within the preceding 2 years) history thereof. Consistent with being in good health
9	Migraine disorder not considered part of the ME/CFS condition (at least one attack in the last 24 months) that may impact study assessments. To avoid interference with the purpose of the study	Migraine (at least one attack in the last 24 months). Consistent with being in good health

Continued

Table 2 Continued

	Patients	Controls
	Justification/rationale	Justification/rationale
10	Recurrent headache or back pain not attributable to ME/CFS occurring on more than 5 days per month in the last 3 months. To avoid interference with the purpose of the study	Recurrent headache or back pain on more than 5 days/month in the last 3 months. Consistent with being in good health
11	Active internal disease or clinically significant sequelae of internal disease (eg, diabetes mellitus, liver diseases, kidney diseases, cardiovascular diseases, hypo- or hyperthyroidism, hypertension, etc) not related to ME/CFS that may impact the conduct or interpretation of study assessments. To avoid interference with the purpose of the study	Active internal disease or sequelae of internal disease (eg, diabetes mellitus, liver diseases, kidney diseases, cardiovascular diseases, hypo- or hyperthyroidism, hypertension, etc). Consistent with being in good health
12	Clinically significant disease (eg, medical history of infection with HIV Type 1 or Type 2, hepatitis B or hepatitis C) or condition that may affect efficacy or safety assessments, or any other reasons, which, in the investigator's opinion, may preclude the subject's participation in the trial. Safety of investigator and their staff Standardisation of the trial population	Clinically significant disease (eg, medical history of infection with HIV Type 1 or Type 2, hepatitis B or hepatitis C) or condition that may affect efficacy or safety assessments, or any other reasons, which, in the investigator's opinion, may preclude the subject's participation in the trial. Safety of investigator and their staff Standardisation of the trial population
13	Not willing or able to abstain from alcohol from 24 hours prior to any study visit and until the end of the study. To avoid interference with the purpose of the study	Not willing or able to abstain from alcohol from 24 hours prior to the study. To avoid interference with the purpose of the study
14	Consumption of cannabis in the last 4 weeks prior to the study. To avoid interference with the purpose of the study	Consumption of cannabis in the last 4 weeks prior to the study. To avoid interference with the purpose of the study
15	Habitually smoking more than 10 cigarettes, 2 cigars or 2 pipes of tobacco per day within the last 6 months before enrolment in this trial. To avoid interference with the purpose of the study	Habitually smoking more than 10 cigarettes, 2 cigars or 2 pipes of tobacco per day within the last 6 months before enrolment in this trial. To avoid interference with the purpose of the study
16	Known or suspected of not being willing or able to comply with the requirements of the trial protocol or the instructions. To ascertain the subject's suitability for the study	Known or suspected of not being willing or able to comply with the requirements of the trial protocol or the instructions. To ascertain the subject's suitability for the study
17	Inability to communicate meaningfully with the trial site staff (eg, insufficient language skills). To ascertain the subject's safety	Inability to communicate meaningfully with the trial site staff (eg, insufficient language skills). To ascertain the subject's safety
18	Any person with direct involvement in the trial conduct; any person under the direct supervision of the investigator or dependent on the investigator. Ethical requirement	Any person with direct involvement in the trial conduct; any person under the direct supervision of the investigator or dependent on the investigator. Ethical requirement
19	Blood loss of 500 mL or more (eg, owing to blood donation) within 3 months before enrolment in this trial. To ascertain the subject's suitability for the study	Blood loss of 500 mL or more (eg, owing to blood donation) within 3 months before enrolment in this trial. To ascertain the subject's suitability for the study

Continued

Table 2 Continued

	Patients	Controls
	Justification/rationale	Justification/rationale
20	Pregnancy, planned pregnancy or lactation. Ethical requirement to protect the unborn or newborn child	Pregnancy, planned pregnancy or lactation. To ensure alignment with the patient group
21	Participation in a structured graded exercise therapy (GET) programme To avoid confounding effects on post-exertional malaise and other ME/CFS symptoms.	Participation in a structured exercise programme To ensure consistency across study arms and avoid potential confounding factors.
A set of <i>temporary exclusion criteria</i> have also been defined. The subject is not excluded if some of these temporary exclusion criteria are met. Instead, the first study visit may be postponed. Before the start of the first study visit, previously met temporary exclusion criteria will be checked again, and their absence will be verified before the screening for study visit one takes place. In addition, to minimise acute physiological confounding, participants will be instructed to refrain from consumption or use of caffeine, alcohol and tobacco, and from intake of increased amounts of salt or fluids, for at least 24 hours prior to baseline and post-treatment assessment visits, as well as before each treatment session. Compliance with these pre-visit restrictions will be verified at each visit, and visits may be postponed if the restrictions are not met.		
22	N/A	Any transient illness within 2 weeks before screening. To avoid interference with the purpose of the study
23	Significant changes in physical exercise activities, for example, starting strenuous workout/training within 1 week before screening. To avoid interference with the purpose of the study	Significant changes in physical exercise activities, for example, starting strenuous workout/training within 1 week before screening. To avoid interference with the purpose of the study
24	N/A	Current or recent (during the preceding 2 weeks) acute pain lasting more than 4 hours. To avoid interference with the purpose of the study
IHHT, intermittent hypoxia-hyperoxia treatment; ME/CFS, myalgic encephalomyelitis/chronic fatigue syndrome.		

Table 3 Exclusion criteria at treatment visits

	Patients	Controls
	Justification/rationale	Justification/rationale
25	For female subjects of childbearing potential: positive or missing pregnancy test To protect the fetus	For female subjects of childbearing potential: positive or missing pregnancy test To ensure alignment with the patient group
26	Blood loss of 500 mL or more (eg, owing to blood donation) since screening. To ascertain the subject's suitability for the study	Blood loss of 500 mL or more (eg, owing to blood donation) since screening. To ascertain the subject's suitability for the study
Temporary exclusion criteria at treatment visits. The subject is not excluded if some of these temporary exclusion criteria are met at screening of the treatment visit. Instead, the treatment visit may be postponed. If this is the case, all temporary exclusion criteria will be checked again.		
27	Alcohol consumption in the last 24 hours prior to the study visit. Subject safety	Alcohol consumption in the last 24 hours prior to the study visit. Subject safety
28	Incidentally not willing or able to comply with the requirements of the trial protocol or the instructions or to communicate meaningfully with the trial site staff. To ascertain the subject's suitability for the study	Incidentally not willing or able to comply with the requirements of the trial protocol or the instructions or to communicate meaningfully with the trial site staff. To ascertain the subject's suitability for the study

fatigue-related disorders.^{47 48} A validated Danish version of the FSS is available and has shown high reliability and construct validity in Danish patients with systemic lupus erythematosus.⁴⁸

Given the central role of fatigue in ME/CFS, the FSS provides a clinically relevant and widely used measure of fatigue severity and its impact on daily functioning; however, it is limited by ceiling effects, as patients typically exhibit uniformly high scores.

COMPASS-31

The COMPASS-31 is a validated self-administered questionnaire developed to quantify autonomic symptom burden across six domains: orthostatic intolerance (four items), vasomotor (three items), secretomotor (four items), gastrointestinal (12 items), bladder (three items) and pupillomotor (two items).⁴⁹ Each item is scored and weighted according to domain-specific algorithms, generating a total score ranging from 0 to 100, with higher scores indicating more severe autonomic dysfunction.

The questionnaire captures a wide spectrum of autonomic symptoms, including dizziness or presyncope, altered sweating, gastrointestinal dysmotility, urinary disturbances and pupillary or thermoregulatory abnormalities. The COMPASS-31 demonstrates excellent internal consistency (Cronbach's $\alpha > 0.9$), strong test-retest reliability and sensitivity to clinical change in conditions characterised by autonomic dysregulation such as postural orthostatic tachycardia syndrome (POTS), diabetic autonomic neuropathy and ME/CFS.^{49–51} This study will use the validated Danish translation of COMPASS-31 developed and linguistically validated by Brinith *et al*,⁵² ensuring semantic and conceptual equivalence with the original instrument.

FUNCAP-27 and PEM

The Functional Capacity questionnaire, 27-item version (FUNCAP-27), is a patient-informed, self-administered questionnaire developed to assess functional capacity and PEM in individuals with ME/CFS.⁵³ The instrument is designed to capture activity-related limitations and the consequences of exertion across daily physical, cognitive and sensory activities. Unlike traditional activity questionnaires that focus on activity frequency or duration, FUNCAP-27 evaluates the severity and duration of post-exertional consequences and the extent to which undertaking a specific activity limits the ability to perform other activities, thereby directly addressing PEM.

FUNCAP-27 comprises 27 items distributed across eight functional domains: personal hygiene/basic functions, walking and movement, being upright, activities in the home, communication, activities outside the home, reactions to light and sound and concentration. Each item is scored on a 0–6 ordinal scale reflecting the impact of the activity on subsequent functional capacity, with higher scores indicating greater functional capacity. Participants are instructed to respond with reference to an average

day during the preceding month, in accordance with the original instrument instructions.

The questionnaire was developed through multiple rounds of structured patient feedback and large-scale national and international surveys and has demonstrated strong internal consistency (Cronbach's α for total score approximately 0.95), high test-retest reliability and clear discrimination across ME/CFS severity categories.⁵³ FUNCAP-27 has been proposed as particularly suitable for interventional research and has also been suggested for use in conditions characterised by fatigue and PEM, including long COVID. In the present study, FUNCAP-27 will be used to assess changes in functional capacity and PEM in response to IHHT.

Pain assessment (painDETECT)

Pain characteristics will be assessed using the painDETECT questionnaire, a validated, self-administered instrument designed to capture pain qualities, sensory symptom patterns and pain distribution in individuals with chronic pain.⁵⁴ The questionnaire includes items assessing pain intensity, pain course pattern, pain radiation and sensory descriptors such as burning sensations, tingling or prickling, electric shock-like pain, thermal sensitivity, numbness, pressure-evoked pain and allodynia. Responses are combined into a composite score reflecting the overall burden and profile of pain-related sensory symptoms.

In the present study, painDETECT will be used descriptively to characterise pain phenotypes and to explore changes in pain-related sensory symptoms following IHHT. A Danish-language version previously applied in Danish clinical pain cohorts was used.⁵⁵

Cognitive performance: d2-R test of attention

Cognitive performance will be assessed using the d2-R Test of Attention – Revised, a validated and widely used measure of selective and sustained attention, processing speed and concentration capacity. The d2-R is a visual cancellation task requiring rapid discrimination of target stimuli while inhibiting responses to visually similar distractors, thereby providing an objective assessment of attentional control and mental processing efficiency.⁵⁶ Deficits in attention, processing speed and concentration are commonly reported in ME/CFS and are considered key contributors to cognitive fatigue and the symptom complex often referred to as ‘brain fog’.^{57–59} The test consists of 14 timed segments, each lasting 20s, during which participants scan rows of symbols and mark predefined target items as quickly and accurately as possible. Primary outcome measures include concentration performance (correct responses minus commission errors), total processed targets, omission errors and commission errors. Test results are interpreted using age- and sex-adjusted normative data derived from large population samples. The d2-R has demonstrated good reliability, construct validity and sensitivity to attentional impairments across clinical and non-clinical populations.^{56 60}

Physical performance – hand-grip strength and muscle fatigability

Muscle strength and fatigability: handgrip strength test

Handgrip strength will be assessed as an objective, low-burden measure of muscle strength, fatigability and post-exertional impairment in ME/CFS. Reduced handgrip strength and increased performance fatigability have consistently been reported in ME/CFS and have been shown to discriminate patients from healthy controls and to correlate with disease severity, physical functioning and PEM.^{11 61 62}

Participants will perform repeated maximal voluntary contractions using a calibrated hand dynamometer while seated in an upright position with the forearm supported. After familiarisation, participants will complete a series of repeated maximal grips with standardised rest intervals, allowing assessment of peak force, mean force across repetitions and decline in force over time. Repeated-measures handgrip protocols have been shown to capture exertion-related performance decline and impaired recovery in ME/CFS, even in the absence of maximal whole-body exercise.⁶³

Handgrip-derived fatigability indices have been associated with PEM severity, muscle pain, autonomic dysfunction and biochemical markers of exertional stress in ME/CFS, supporting their relevance as both functional and mechanistic outcome measures.^{61 63 64}

Tissue oxygenation and microvascular function

NIRS during handgrip exercise

NIRS will be used to assess skeletal muscle oxygenation, haemodynamic responses, both at rest, during handgrip exercise and after handgrip exercise during occlusion to measure microvascular responsiveness. NIRS is a non-invasive optical technique that enables continuous, real-time assessment of relative changes in oxygenated haemoglobin, deoxygenated haemoglobin and total haemoglobin, providing indices of local oxygen delivery, extraction and microvascular perfusion.^{65 66}

In ME/CFS, impaired oxygen delivery, altered microvascular regulation and abnormal muscle oxygen utilisation during physical activity have been proposed to contribute to early fatigability and post-exertional symptom exacerbation. Studies combining submaximal exercise with NIRS have demonstrated reduced oxygen extraction, blunted haemodynamic responses and delayed recovery kinetics in ME/CFS compared with healthy controls.⁶⁷ There is increasing evidence for disturbed vascular responsiveness and peripheral endothelial dysfunction in ME/CFS, which relates to disease severity.^{24 68 69} In a case-control study, Miller *et al* showed that NIRS measurements obtained during repetitive submaximal handgrip exercise revealed abnormal patterns of muscle oxygenation and impaired recovery in ME/CFS.⁶⁷

In the present study, NIRS will be recorded continuously during handgrip strength testing, at baseline and post-treatment and during occlusion to measure microvascular responsiveness. This integrated approach allows

simultaneous assessment of mechanical performance and muscle oxygenation dynamics, enabling evaluation of exertion-related changes in oxygen utilisation and recovery. NIRS-derived outcomes will be explored as physiological correlates of fatigability and PEM and as mechanistic endpoints responsive to IHHT.

Autonomic function testing

Autonomic cardiovascular regulation will be assessed using a combination of active orthostatic challenge (NLT) and autonomic reflex testing (Valsalva manoeuvre). These complementary tests allow evaluation of heart rate and blood pressure regulation during orthostatic stress and assessment of sympathetic and parasympathetic baroreflex function, both of which are frequently impaired in ME/CFS.^{70 71}

NLT

Orthostatic cardiovascular responses will be evaluated using the 10 min NLT. The NLT was originally developed by the National Aeronautics and Space Administration to assess cardiovascular responses to orthostatic stress and has been adapted for clinical use as a practical alternative to head-up tilt table testing.^{72 73} The test has demonstrated good agreement with tilt testing for identifying orthostatic haemodynamic abnormalities, including POTS and orthostatic hypotension.^{73 74}

Following a 10-min supine rest period to establish baseline values of heart rate and blood pressure, participants will assume an upright posture with the heels positioned approximately 15–20 cm from a wall and the shoulders supported against the wall to minimise skeletal muscle pumping. Heart rate and blood pressure will be recorded continuously during the subsequent 10 min of upright posture. The test will be terminated prematurely if participants develop significant symptoms such as dizziness, palpitations, presyncope or cognitive disturbance, consistent with consensus recommendations.^{75 76}

Beat-to-beat blood pressure will be measured non-invasively using finger photoplethysmography based on the volume-clamp technique, allowing continuous assessment of systolic and diastolic blood pressure and heart rate.⁷⁷ Mean heart rate, systolic blood pressure and diastolic blood pressure will be derived from the final 3 min of the supine phase and the final 3 min of the upright phase, as commonly applied in orthostatic testing protocols.^{74 76}

The NLT enables classification of orthostatic haemodynamic responses, including normal responses, POTS, delayed orthostatic hypotension and other forms of orthostatic intolerance. Importantly, orthostatic symptoms and physiological impairment in ME/CFS may occur even in the absence of diagnostic heart rate or blood pressure thresholds, reflecting impaired cerebral perfusion and autonomic dysregulation during upright posture.^{29 78 79}

Valsalva manoeuvre

Autonomic reflex function will be further assessed using the Valsalva manoeuvre, which provides established indices of sympathetic vasomotor function, parasympathetic cardiac control and baroreflex sensitivity.^{75 76 80}

Participants will be tested in a seated position with continuous recording of heart rate and blood pressure. After a maximal inspiration, participants will exhale forcefully into a mouthpiece connected to a manometer, maintaining an expiratory pressure of 40 mmHg for 15 s, followed by normal breathing. The manoeuvre will be repeated up to three times, and the technically optimal recording will be selected for analysis, in accordance with standard autonomic testing recommendations.^{76 80}

The characteristic four phases of the Valsalva manoeuvre will be analysed. During phase II, reductions in venous return and stroke volume provoke compensatory sympathetic vasoconstriction, reflected in recovery of blood pressure (phase IIb). Following release, the transient blood pressure overshoot (phase IV) and associated heart rate slowing reflect parasympathetic reactivation.^{76 80}

Derived outcome measures will include: (i) the slope of systolic blood pressure recovery during phase IIb as an index of sympathetic vasomotor function; (ii) baroreflex sensitivity, calculated from the relationship between RR interval changes and systolic blood pressure following release; (iii) the Valsalva ratio (maximum heart rate during strain divided by minimum heart rate within 30 s after release); and (iv) indices reflecting sympathovagal balance. Abnormalities in these measures have been reported in ME/CFS and related autonomic disorders, consistent with impaired baroreflex control, altered sympathetic activation and reduced parasympathetic modulation.^{29 75}

Sudomotor function assessment (udoscan)

Sudomotor autonomic function will be assessed using Sudoscan (Impeto Medical, Paris, France), a non-invasive device designed to evaluate sweat gland function and small-fibre autonomic nerve integrity through measurement of electrochemical skin conductance (ESC) in the hands and feet. Sudomotor dysfunction is considered a marker of small-fibre autonomic neuropathy and has been associated with autonomic dysfunction in neurological and metabolic disorders.^{81 82}

During the assessment, participants will place their palms and soles on stainless steel electrodes, and a low-voltage electrochemical stimulus will be applied to measure chloride ion conductance generated by sweat glands. ESC values will be recorded separately for hands and feet, together with asymmetry measures between sides. Reduced ESC values may reflect impaired small-fibre autonomic function.

The examination is non-invasive, painless and completed within approximately 2 min. Sudoscan measurements will be performed at baseline and post-treatment as exploratory markers of peripheral autonomic and small-fibre dysfunction and will be analysed in relation to clinical

symptoms, autonomic testing and other neurophysiological measures.

Thermal QST

Thermal quantitative sensory testing (QST) will be performed to assess thermal detection and pain thresholds mediated by small sensory nerve fibres, including thinly myelinated Aδ fibres and unmyelinated C fibres. QST provides a standardised, non-invasive psychophysical method for evaluating somatosensory function and is widely used to characterise small-fibre dysfunction in neuropathic pain conditions and fibromyalgia.^{83–85}

Thermal QST will be conducted using a computer-controlled NeuroSensory Analyzer (TSAIL, Medoc Ltd., Israel) equipped with a Peltier-based thermode (30×30 mm). Testing will be performed in a quiet, temperature-controlled room (22–24 °C) following a standardised acclimatisation period to minimise environmental influences on sensory perception. Participants will receive uniform verbal instructions and will be informed that they remain in control of the procedure and may terminate testing at any time.

Cold detection threshold (CDT) and warm detection threshold (WDT) will be assessed starting from a baseline temperature of 32 °C. For CDT, temperature will decrease at a rate of 1 °C/s, and participants will indicate the first perception of cold. For WDT, temperature will increase at 1 °C/s, and participants will indicate the first perception of warmth. Each modality will be repeated three times, and mean threshold values will be calculated. CDT and WDT primarily reflect the function of Aδ and C fibres, respectively, and provide sensitive indices of small-fibre sensory integrity.^{84 85}

Cold pain threshold and heat pain threshold will be assessed using similar ramped stimuli, with temperature decreasing or increasing from baseline until the participant reports the first sensation perceived as painful. Participants will be instructed to respond at the transition from non-painful to painful sensation and not to tolerate intolerable pain. Safety cut-off limits will be applied in accordance with established guidelines to prevent tissue injury. Each pain threshold will be assessed three times, and mean values will be used for analysis. Thermal pain thresholds predominantly reflect C-fibre-mediated nociceptive processing.^{83 85 86}

Thermal QST will be performed at predefined anatomical sites selected according to the study protocol. Skin will be inspected prior to testing to exclude areas with lesions, inflammation or altered integrity. Between stimuli, sufficient time will be allowed for skin temperature to return to baseline. Raw threshold values and averaged measures will be recorded, and tests with inconsistent responses or delayed reactions will be flagged as unreliable. Where appropriate, results will be interpreted in relation to established normative reference data.^{84 87}

Thermal QST has demonstrated good test–retest reliability, sensitivity to small-fibre pathology and clinical relevance across a range of chronic pain and fatigue-related

disorders. Altered thermal detection and pain thresholds may reflect changes in sensory processing.^{85 88 89}

Skin biopsy and histological analyses

Skin punch biopsies will be obtained to assess cutaneous small-fibre nerve integrity, neuroimmune features, mitochondrial markers and microvascular structures. Skin biopsy is a minimally invasive and well-validated technique for investigating peripheral mechanisms contributing to pain, sensory dysfunction and autonomic symptoms and is widely used in the evaluation of small-fibre neuropathy and chronic pain disorders.^{89–92}

A 3-mm punch skin biopsy will be obtained from the lower leg of each participant, 10 cm proximal to the lateral malleolus, under aseptic conditions and local anaesthesia using 1% lidocaine. Biopsies will be performed by trained personnel according to established safety standards. Following the biopsy sampling, the biopsy site will be gently compressed to achieve haemostasis and covered with a sterile adhesive dressing.^{90 91}

Biopsy specimens will be processed according to standard protocols for immunohistochemical analysis. Tissue will be fixed, cryoprotected and sectioned for staining with antibodies against the pan-axonal marker protein gene product 9.5 (PGP9.5). Intraepidermal nerve fibre density (IENFD) will be quantified using established stereological and spatial sampling methods, expressed as the number or length of nerve fibres crossing the dermal-epidermal junction per unit length of epidermis, in accordance with published guidelines and validated analytical approaches.^{90 93 94}

To characterise autonomic nerve fibres within the skin, immunohistochemical staining of selected dermal structures will include markers of cholinergic and adrenergic innervation. Vasoactive intestinal peptide (VIP) will be used as a marker of predominantly parasympathetic cholinergic fibres, particularly within autonomic target structures such as sweat glands and pilomotor units, where VIP-positive fibres reliably reflect autonomic innervation rather than sensory fibres.^{91 95 96} Dopamine β -hydroxylase (DbH) will be used to identify sympathetic adrenergic fibres, as it is selectively expressed in norepinephrine-producing neurons and is a well-established marker of sympathetic autonomic innervation in human skin.⁹⁷ These markers will be interpreted in the context of their known expression patterns and anatomical localisation and, where appropriate, analysed alongside the pan-neuronal marker PGP9.5 and structural markers to distinguish autonomic from sensory nerve fibres, in accordance with established skin biopsy guidelines.^{90 98}

To characterise non-neuronal features, additional immunohistochemical staining will be performed to identify immune cells (including macrophages), mitochondrial markers and vascular structures within the dermis. Quantitative and semi-quantitative analyses will be conducted in a blinded manner using standardised image analysis procedures.

IENFD is a sensitive structural measure of small-fibre pathology and is widely used in the evaluation of neuropathies. Reductions in IENFD have been reported across a range of neuropathic and inflammatory conditions and provide an objective measure of damage to sensory nerve fibres.^{90 94 99} In ME/CFS and fibromyalgia, previous studies have reported reduced IENFD, altered nerve morphology and evidence of peripheral nerve involvement, supporting a contribution of cutaneous small-fibre pathology to disease mechanisms and symptom expression.^{81 89 92 100} Notably, abnormalities detected in upper extremities using Sudoscan suggest that small-fibre pathology in ME/CFS may be non-length-dependent in a subset of patients.⁸¹ Consistent with this heterogeneity, reduced IENFD compatible with small-fibre neuropathy has been reported in approximately one-third of patients with ME/CFS.²⁷

Beyond nerve fibre density, skin biopsy allows investigation of peripheral neuroimmune interactions, which are increasingly recognised as contributors to chronic pain and sensory hypersensitivity. Immune cell-nerve interactions within the skin may modulate nociceptor and autonomic fibre excitability and contribute to persistent pain, fatigue and post-exertional symptom exacerbation.^{101 102} These mechanisms are of particular relevance in ME/CFS, where immune dysregulation, fatigue, pain and exertional intolerance frequently coexist.

Skin biopsies will be obtained at baseline and post-treatment to explore whether IHHT is associated with changes in cutaneous nerve integrity, immune and mitochondrial features or vascular markers, and to relate these findings to clinical outcomes, physiological measures and patient-reported symptoms.

CCM

CCM will be performed to assess small-fibre nerve morphology in vivo by imaging the corneal subbasal nerve plexus. CCM is a non-invasive technique that provides quantitative measures of corneal nerve structure and is a validated surrogate marker of peripheral small-fibre pathology.^{103–105} CCM will be performed using a Heidelberg Retina Tomograph III with the Rostock Cornea Module (Heidelberg Engineering, Germany). Following instillation of topical local anaesthetic eye drops, high-resolution images of the central corneal sub-basal nerve plexus will be acquired from each eye. The examination is brief and well tolerated.

Images will be analysed in a masked manner using validated image-analysis software. Quantitative outcome measures will include corneal nerve fibre density, corneal nerve fibre length and corneal nerve branch density. These parameters have demonstrated good reproducibility and sensitivity for detecting small-fibre nerve damage and regeneration and have been shown to correlate with IENFD density assessed by skin biopsy.^{104–107}

CCM will be performed at baseline and post-treatment as an exploratory outcome to assess potential changes in small-fibre nerve morphology and to complement

thermal QST and skin biopsy findings. Small-fibre abnormalities may contribute to autonomic dysfunction, tissue microcirculation/oxygenation and overall symptom amplification.

Biobank and molecular measures

IHHT is known to activate hypoxia- and redox-sensitive molecular pathways and to induce systemic immune-metabolic adaptations that can be captured in peripheral blood and immune cell populations.^{33–35} To investigate molecular signatures associated with ME/CFS and potential treatment-related biological changes following IHHT, a structured biobanking strategy will be implemented, enabling both targeted and exploratory mechanistic analyses.

Venous blood samples will be collected at baseline and at the post-treatment follow-up assessment. Plasma and peripheral blood mononuclear cells (PBMCs) will be isolated and stored under standardised conditions for subsequent molecular analyses. Targeted analyses will include markers of oxidative stress, such as lipid peroxidation products and isoprostanes, and inflammatory and vascular mediators including cytokines and chemokines, quantified using ELISA-based methods. In addition, unbiased profiling approaches will be applied to assess global changes in circulating molecular signatures related to cellular stress responses, metabolic energy homeostasis and immune signalling, using metabolomic, proteomic and transcriptomic analyses. These approaches are well established for capturing hypoxia- and inflammation-related systemic adaptations in ME/CFS and post-viral fatigue syndromes.^{12 21 22 35}

Functional bioenergetic profiling of PBMCs will be performed using extracellular flux analysis to quantify ATP production derived from mitochondrial oxidative phosphorylation and glycolysis. Complementary gene and protein expression analyses will focus on redox- and oxygen-sensing pathways, including transcriptional regulators, mitochondrial energy metabolism markers, and cellular stress and repair proteins. Immune-metabolic dysregulation at the PBMC level has been repeatedly demonstrated in ME/CFS and is considered a key mechanistic contributor to PEM and systemic symptom burden.^{11 25 33}

Randomisation and blinding

Participants will be randomised in a 1:1 ratio to the intervention or placebo group using a computer-generated randomisation sequence implemented within the REDCap randomisation module. Randomisation will occur after completion of the baseline assessment visit and confirmation of eligibility, ensuring that all inclusion and exclusion criteria are met prior to allocation.

The allocation sequence will be generated externally using a secure randomisation service to create a permuted block randomisation scheme, with predefined block sizes to ensure balanced group allocation throughout the recruitment period. Randomisation will be stratified by

predefined variables, including age, disease duration and disease severity as assessed by the SF-36 vitality score. The resulting allocation table will be uploaded into REDCap by a trusted staff member who is not involved in participant recruitment, assessment or treatment, thereby ensuring allocation concealment.

Within REDCap, the randomisation module will assign participants automatically based on the uploaded allocation table and stratification variables. Investigators and recruiting staff will have no access to the allocation sequence and will be unable to predict upcoming assignments.

The study will be conducted using a blinded randomisation setup. Treatment allocation will be concealed using unique randomisation codes, and the mapping between codes and treatment groups will be stored securely by designated personnel not otherwise involved in the trial. Participants, investigators, outcome assessors and data analysts will remain blinded to treatment allocation throughout the study. To evaluate blinding success, participants will be asked to guess their treatment allocation after completion of the intervention. These data will be analysed descriptively and incorporated into sensitivity analyses to assess the potential impact of potential unblinding on study outcomes. Procedures for emergency unblinding will be predefined and implemented only if clinically required, in accordance with GCP principles.

This randomisation and blinding procedure follows institutional guidelines for REDCap-based randomisation in clinical trials and ensures robust allocation concealment and protection against selection and performance bias.

Safety and AEs

The intervention is considered low risk, and IHHT administered according to the study protocol has been shown to be safe and well tolerated in previous studies.^{33 40} No permanent or severe adverse effects are known to be associated with the applied treatment regimen.

Expected transient symptoms associated with IHHT include headache on the day of or the day following treatment and short-lasting fatigue, which may persist for up to several days after a session. Mild and transient fatigue following treatment may represent an anticipated physiological response to the intervention and will be monitored throughout the study period. However, clinically significant symptom exacerbations consistent with PEM, including prolonged worsening of symptoms persisting beyond 48 hours and substantial functional decline, will be prospectively documented and assessed as adverse events (AEs). All other AEs will be recorded and evaluated according to predefined criteria.

Risks associated with study-related procedures are minimal. Blood sampling may cause transient discomfort, bruising, minor bleeding or swelling at the puncture site and rarely local infection. Skin punch biopsy may be associated with mild pain, minor bleeding and, in rare cases, local infection. These procedures are routinely

performed at the study sites, and complications are uncommon and typically self-limiting.

Participant safety will be ensured through continuous monitoring during treatment sessions, including assessment of oxygen saturation, heart rate and clinical signs of discomfort. Treatment parameters will be individually adjusted to ensure tolerability. Blood sampling and skin biopsies will be performed by trained personnel using sterile techniques, and participants will receive appropriate aftercare instructions. Medical staff will be available throughout the study period to address any AEs or concerns. All AEs and serious AEs will be documented and reported in accordance with GCP guidelines.

Trial monitoring

Trial conduct will be monitored by a project nurse independent of the investigators responsible for treatment and data collection, using regular monitoring visits and remote monitoring. Monitoring will focus on informed consent, protocol adherence and source data verification for key variables and AE reporting. The monitor will report findings to the sponsor.

No independent data monitoring committee has been established due to the low-risk nature of the intervention and continuous monitoring during IHHT sessions.

Data management

All study data will be collected and managed using REDCap electronic data capture tools hosted at Aarhus University. Built-in range checks, validation rules and audit trails will be used to promote data quality. Access to identifiable data will be restricted to authorised study personnel, and data will be stored securely in accordance with the General Data Protection Regulation (GDPR) and institutional policies.

To minimise potential bias and ensure scientific independence, investigators affiliated with SANA Medical Systems ApS and Remedical.Me ApS will not have access to unblinded data prior to database lock and will not be involved in the final statistical analyses or interpretation of efficacy outcomes. Data analysis will be conducted by blinded investigators independent of the industry collaborators.

Statistical analysis

General principles and analysis populations

All analyses will be prespecified in a statistical analysis plan finalised prior to database lock. The primary analysis will follow the intention-to-treat principle, including all randomised participants analysed according to assigned treatment group. A per-protocol sensitivity analysis will also be conducted including participants who complete the intervention with no major protocol deviations.

Continuous variables will be summarised as mean (SD) or median (IQR) as appropriate and categorical variables as counts and percentages. All statistical tests will be two-sided with a significance level of 0.05. Analyses

will be performed using R (R Foundation for Statistical Computing).

Primary endpoint analysis

The primary endpoint is the change from baseline to post-treatment in the SF-36 vitality domain score. The primary comparison between intervention and placebo will be performed using an analysis of covariance (ANCOVA) model with post-treatment SF-36 vitality as the dependent variable, treatment group as the main predictor, and baseline SF-36 vitality as a covariate. Stratification variables used in randomisation (age, disease duration and disease severity) will be included as covariates where appropriate.

Results will be reported as adjusted mean differences between groups with 95% CIs and corresponding p values.

No interim analyses or formal stopping guidelines are planned. In the event of unexpected safety concerns, the sponsor and medically responsible investigator have the authority to pause or terminate the trial.

Secondary endpoints

Secondary endpoints (including other SF-36 domains, FSS, FUNCAP-27, COMPASS-31 and objective performance measures) will be analysed using ANCOVA models analogous to the primary analysis (post-treatment outcome adjusted for baseline) or appropriate generalised models for non-normally distributed data. Where outcomes are measured at multiple time points, mixed-effects models for repeated measures (MMRM) will be used to estimate group differences over time, including fixed effects for treatment group, time and group-by-time interaction, and adjusting for baseline values.

Secondary endpoint analyses will be interpreted with attention to multiplicity. The primary endpoint will be tested confirmatorily; secondary and exploratory outcomes will be interpreted as supportive and hypothesis-generating unless otherwise prespecified.

Long-term follow-up PROMs (3, 6 and 12 months)

Online PROMs collected at 3, 6 and 12 months post-intervention will be analysed using MMRM to evaluate durability of treatment effects over time. Estimated marginal means with 95% CIs will be reported for each time point.

Exploratory mechanistic endpoints

Exploratory physiological and mechanistic measures (eg, tissue oxygenation, autonomic sudomotor indices, CCM, skin biopsy measures and molecular biomarkers) will be analysed descriptively and using regression or mixed-effects models as appropriate. For high-dimensional molecular analyses (eg, RNA sequencing or untargeted mass spectrometry), analyses will follow established exploratory pipelines. Where appropriate, adjustment for multiple testing (eg, false discovery rate control) may be applied. Exploratory analyses will examine whether baseline autonomic, small-fibre, vascular or immune-metabolic phenotypes are associated with treatment response to generate hypotheses about responder profiles.

Healthy control group

A healthy control group will be included for baseline comparisons. Controls will not be included in the randomised treatment effect analyses. Baseline comparisons between patients and controls will be performed using appropriate two-sample tests (t-test/Wilcoxon for continuous variables; χ^2 /Fisher's exact for categorical variables), and effect sizes with CIs will be reported.

Missing data

Every effort will be made to minimise missing data. Primary analyses will use methods appropriate to the model (ANCOVA/MMRM) under a missing-at-random assumption. Sensitivity analyses may include multiple imputation depending on the extent and nature of missingness.

Sample size

The sample size is based on the primary endpoint, defined as the mean change from baseline to post-treatment in the SF-36 vitality score. Based on data from a non-controlled IHHT cohort of long-COVID patients, a mean improvement of 20.5 points with a SD of 22.5 was observed.⁴⁰ Assuming a placebo effect corresponding to a 33% improvement (6.8 points), the expected between-group difference is 13.7 points.

With a two-sided alpha level of 0.05 and 80% power, and assuming equal SDs in both groups, the sample-size calculation using an independent samples t-test indicates that 52 participants per group would be required, including an anticipated dropout rate of 15%, corresponding to a total target sample size of 104 patients. In addition, 20 healthy controls will be included for baseline reference measurements only.

DISCUSSION

This article presents the design and scientific rationale for a randomised, placebo-controlled trial investigating IHHT in patients with ME/CFS. The trial is conducted in accordance with ICH-GCP guidelines to safeguard participant safety and to ensure the integrity and validity of the collected data.

IHHT has been increasingly applied as a non-pharmacological intervention aimed at inducing adaptive responses in mitochondrial function, vascular regulation, immune-inflammatory signalling and systemic metabolic resilience and has demonstrated a favourable safety and tolerability profile across a range of clinical and preventive settings.³³ Nevertheless, the intervention is not without transient physiological effects. Given the multisystem symptom burden, marked functional impairment, and reduced quality of life associated with ME/CFS, the study population represents a potentially vulnerable group.⁴³ Particular emphasis has therefore been placed on thorough informed consent procedures, expectation management and careful monitoring throughout the trial to minimise undue influence or therapeutic misconception.

A major strength of the present study is its randomised, participant- and assessor-blinded design, which represents the methodological gold standard for evaluating patient-reported outcomes. This is particularly relevant in ME/CFS, where subjective symptoms such as fatigue and PEM constitute core disease features. The inclusion of a placebo intervention and rigorous blinding of participants and investigators is intended to reduce bias and enhance the interpretability of clinical efficacy outcomes. In addition, inclusion of a well-characterised healthy control group allows physiological, autonomic and molecular measures to be interpreted relative to normative values, strengthening inferences regarding disease-specific mechanisms.

In addition to the primary and secondary efficacy endpoints, the trial incorporates a broad range of exploratory and descriptive outcome measures. These include assessments of autonomic function, physical and cognitive performance, tissue oxygenation, peripheral nerve integrity and immune-metabolic and oxidative stress biomarkers. This comprehensive phenotyping strategy may not only generate hypotheses about responder profiles but may also provide valuable information on disease mechanisms and heterogeneity, independent of clinical efficacy outcomes.

Should IHHT demonstrate beneficial effects on quality of life, fatigue or functional capacity, the study design allows for parallel investigation of potential underlying mechanisms, including microvascular function, autonomic regulation, mitochondrial bioenergetics and immune-metabolic signalling. Conversely, even in the absence of clear clinical benefit, the collected physiological and molecular data are expected to contribute to a deeper understanding of ME/CFS pathophysiology and to inform future hypothesis-driven studies.

In summary, this trial is designed to rigorously evaluate the safety, feasibility and potential efficacy of IHHT in ME/CFS while simultaneously generating mechanistic data in a condition for which effective treatments remain limited. The results may inform future therapeutic strategies and support the development of more targeted, mechanism-based interventions for ME/CFS and related post-infectious conditions.

LIMITATIONS

The present trial has several important limitations that should be considered when interpreting its findings. ME/CFS is a heterogeneous and incompletely understood condition with complex, multisystem pathophysiology. Although growing evidence implicates immune-metabolic, vascular and autonomic dysfunction, the relative contribution of these mechanisms may vary substantially between individuals. Consequently, treatment effects may differ across patient subgroups, and the trial is not powered to perform definitive subgroup analyses.

The IHHT protocol applied in this study represents a relatively novel intervention in the context of ME/CFS.

While intermittent hypoxia-hyperoxia approaches have demonstrated favourable safety and physiological effects in other populations, the optimal dose, duration and frequency of IHHT for ME/CFS are not yet established. The selected treatment regimen is based on existing clinical experience and feasibility considerations rather than on prior pilot trials in this specific population, which constitutes a limitation. Further studies may be required to refine treatment parameters and to assess longer-term or alternative dosing strategies.

Potential selection bias should also be acknowledged. Participants are recruited from specialised outpatient settings, and inclusion requires a confirmed diagnosis of ME/CFS, which may favour individuals with more severe or longstanding disease. At the same time, individuals with the most severe disease, including those who are bedbound or unable to attend in-person assessments, are unlikely to be represented. As a result, the study population may not be fully representative of the broader ME/CFS spectrum, particularly individuals with milder symptoms, shorter disease duration or very severe functional impairment, potentially limiting generalisability.

The sample size calculation for the primary endpoint is extrapolated from observational data in related post-viral fatigue conditions rather than from a dedicated pilot trial in ME/CFS. Although the study is adequately powered to detect a clinically meaningful difference in the SF-36 vitality domain, assumptions regarding effect size and variability may be imprecise, introducing a risk of type II error. In addition, placebo responses in fatigue-related conditions can be substantial, which may further reduce the detectable between-group difference. Although the sham intervention replicates device-related sensory inputs, it does not reproduce the physiological effects of hypoxia-hyperoxia exposure, and some participants may therefore perceive differences between treatment conditions.

Finally, although the trial includes a comprehensive set of secondary and exploratory physiological and molecular outcomes, it is not powered to detect small effects across these measures. Findings from mechanistic analyses should therefore be interpreted as exploratory and hypothesis-generating rather than confirmatory.

ETHICS AND DISSEMINATION

The trial is conducted in accordance with the Declaration of Helsinki, the ICH-GCP guidelines and applicable Danish and European regulations. The study was approved by the Danish Scientific Ethical Committees (De Videnskabsetiske Medicinske Komitéer, VMK; approval no. 2500959), covering the clinical trial protocol (version 3.0, dated 30 June 2025), participant information and consent materials, questionnaires and the Investigator's Brochure. Any substantial protocol amendments will be communicated to the ethics committee, updated in the trial registry and reported in subsequent publications as appropriate.

Written informed consent will be obtained from all participants prior to any study-related procedures. Participation in optional components, including storage of biological material in a research biobank, requires separate consent and is not a prerequisite for participation in the clinical trial. All data will be handled in compliance with the GDPR and Danish data protection legislation, and biological samples will be stored and managed in accordance with national biobank regulations.

The trial has been registered in a public clinical trials registry prior to participant enrolment (ClinicalTrials.gov: NCT07317401). The study will be monitored in accordance with institutional procedures to ensure compliance with the approved protocol, ICH-GCP and applicable regulatory requirements.

The results of the trial will be disseminated through publication in peer-reviewed scientific journals and presentation at national and international conferences, irrespective of the direction or magnitude of the findings. The primary publication will report the primary and key secondary endpoints together with safety outcomes. Additional publications may report predefined exploratory and mechanistic analyses, including physiological, neurobiological and molecular outcomes.

A summary of the study results will be made available to participants in a format that is accessible and understandable to a non-specialist audience. De-identified data may be shared with other researchers for scientific purposes on reasonable request and in accordance with ethical approvals, data protection regulations and data-sharing agreements.

Participants are covered by the Danish patient compensation system in the event of harm related to trial participation. In addition, participants receiving treatment at SANA are covered by SANA Medical Systems' clinical trial insurance.

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Contributors RKJO was the principal investigator and conceived the study. She led the overall study design and provided scientific oversight throughout protocol development. MN contributed to the study design and drafted the manuscript. SK contributed to drafting the study protocol and the manuscript, including the Introduction. MN and SK share first authorship. NM, LRG, LSB, JM and HT contributed to patient-related aspects of the study, including clinical evaluation procedures and implementation of study procedures. JM additionally contributed to the design and implementation of autonomic testing procedures. MO contributed expertise in mathematical modelling and systems physiology

and provided scientific input on the design of physiological and autonomic assessment components of the study protocol. PK contributed to the design of pain-related outcomes and skin biopsy methodology. CBF and RS contributed to the development and implementation of the IHHT intervention protocol. IG, IMDN and SC contributed to the development, piloting and refinement of assessment protocols and study procedures. All authors contributed to the current submission, critically reviewed the manuscript for important intellectual content, approved the final version and agree to be accountable for all aspects of the work. All authors meet the International Committee of Medical Journal Editors (ICMJE) criteria for authorship, and no individuals meeting these criteria have been omitted. RKJO is the guarantor of the study and accepts full responsibility for the integrity of the work as a whole.

Funding This project is funded by Independent Research Fund Denmark (grant number 3165-00221B, Rikke KJ Olsen) and supported by the Lundbeck Foundation (grant number R359-2020-2620, Nanna Brix Finnerup).

Competing interests LRG discloses employment by and ownership interests in Remedical.Me ApS. CBF and RS are founders of and hold ownership interests in SANA Medical Systems ApS. MN, SK, NM, IG, IMDN, SC, LSB, MO, PK, HT, JM and RKJO declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this manuscript.

Patient and public involvement Patients and/or the public were involved in the design, conduct, reporting or dissemination plans of this research. Refer to the Methods section for further details.

Patient consent for publication Not applicable.

Provenance and peer review Not commissioned; externally peer-reviewed.

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